

2.Robot information release

After the program runs, the ROS expansion board will return real-time data, including voltage, IMU data, and connect to the car encoder to obtain the speed of the car.

1、 Start up

Taking Rosmaster-X3 of our company as an example, enter the following command on the terminal to start it.

```
ros2 run yahboomcar_bringup Mcnamu_driver_X3
```

```
sunrise@ubuntu:~/yahboomcar_ws$ ros2 run yahboomcar_bringup Mcnamu_driver_X3
Rosmaster Serial Opened! Baudrate=115200
X3
imu_link
1.0
1.0
5.0
-----create receive threading-----
```

2、 View topic nodes

Enter the command at the terminal

```
ros2 topic list
```

```
sunrise@ubuntu:~/yahboomcar_ws$ ros2 topic list
/Buzzer
/RGBLight
/cmd_vel
/edition
/imu/data_raw
/imu/mag
/joint_states
/parameter_events
/rosout
/vel_raw
/voltage
sunrise@ubuntu:~/yahboomcar_ws$
```

Topic Name	topical contents
/Buzzer	buzzer
/RGBLight	Light strip effect control
/cmd_vel	Speed control
/edition	Version information
/imu/data_raw	IMU sensor data
/imu/mag	IMU Magnetometer Data
/vel_raw	Car speed information
/voltage	Battery voltage information

3、Reading Topic Data

If you want to read the voltage value, you need to enter the following command at the terminal

```
ros2 topic echo /voltage
```

```
sunrise@ubuntu:~/yahboomcar_ws$ ros2 topic echo /voltage
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
---
data: 12.399999618530273
```

4、Publish topic data

To publish/cmd_ Taking speed data as an example, terminal input

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.5, y: 0.0, z: 0.0},
angular: {x: 0.0, y: 0.0, z: 0.2}}"
```

```

sunrise@ubuntu:~/yahboomcar_ws$ ros2 topic pub /cmd_vel geometry_msgs/msg/Twist
"{linear: {x: 0.5, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.2}}"
publisher: beginning loop
publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.5, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.2))

publishing #2: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.5, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.2))

publishing #3: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.5, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.2))

publishing #4: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=0.5, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=0.2))

```

5、 Source code parsing

```

from Rosmaster_Lib import Rosmaster #Import Driver Library
self.car = Rosmaster() #Instantiating Rosmaster objects
#create subscriber 创建订阅者
self.sub_cmd_vel =
self.create_subscription(Twist,"cmd_vel",self.cmd_vel_callback,1)
self.sub_RGBLight =
self.create_subscription(Int32,"RGBLight",self.RGBLightcallback,100)
self.sub_BUZZer =
self.create_subscription(Bool,"BUZZer",self.BUZZercallback,100)
#create publisher 创建发布者
self.EdiPublisher = self.create_publisher(Float32,"edition",100)
self.volPublisher = self.create_publisher(Float32,"voltage",100)
self.staPublisher = self.create_publisher(JointState,"joint_states",100)
self.velPublisher = self.create_publisher(Twist,"vel_raw",50)
self.imuPublisher = self.create_publisher(Imu,"/imu/data_raw",100)
self.magPublisher = self.create_publisher(MagneticField,"/imu/mag",100)
#调用库，读取ros拓展板的信息
edition.data = self.car.get_version()*1.0
battery.data = self.car.get_battery_voltage()*1.0
ax, ay, az = self.car.get_accelerometer_data()
gx, gy, gz = self.car.get_gyroscope_data()
mx, my, mz = self.car.get_magnetometer_data()
vx, vy, angular = self.car.get_motion_data()
#发布话题数据
self.imuPublisher.publish(imu)
self.magPublisher.publish(mag)
self.volPublisher.publish(battery)
self.EdiPublisher.publish(edition)
self.velPublisher.publish(twist)
#订阅者回调函数
def cmd_vel_callback(self,msg)
def RGBLightcallback(self,msg)
def BUZZercallback(self,msg):

```

Please refer to the following path code for detailed

~/yahboomcar_ws/src/yahboomcar_bringup/yahboomcar_bringup/Mcnamu_driver_X3.py